IoTsmart ESP32-S3 v.2.01

IoTsmart ESP32-S3: The Next Evolution in Connectivity

Discover new possibilities in connectivity and performance with our latest module, meticulously engineered around **Espressif's** cutting-edge **ESP32-S3** microcontroller. This powerful System-on-Chip (SoC) isn't just a component; it's the core of a solution engineered to truly push the boundaries of what's achievable in embedded systems.

The **ESP32-S3** stands out with its **Xtensa® 32-bit LX7** dual-core processor, clocking in at up to 240 MHz, 512KB of SRAM and 384KB ROM, onboard 2MB PSRAM and 4MB Flash memory. This translates to serious processing power, ready to efficiently tackle even your most demanding applications. What truly makes the **ESP32-S3**—and by extension, our module—a transformative force is its robust wireless suite. With integrated 2.4 GHz Wi-Fi (802.11 b/g/n) and Bluetooth® 5 (LE), you're equipped with reliable, versatile communication options right from the start.

Key enhancements in version 2.01 compared to previous versions:

- **Optimized Form Factor**: The module now measures 47x30 mm, providing better mechanical compatibility and a more seamless fit when docked with IoTextra mezzanines.
- **EYESPI Interface**: A new onboard **EYESPI** connector adds plug-and-play support for popular LCDs and other displays using a standardized 18-pin FPC cable.
- **Dual Qwiic© Connectors**: The addition of a second **Qwiic** connector allows for easy daisy-chaining of multiple **I²C** devices and sensors without the need for additional hubs.

The **IoTsmart ESP32-S3** module offers native support for popular IoT platforms:

Node-RED TASMOTA Blynk IoTflow

Comprehensive examples are actively maintained to help accelerate your project development.

Whether you're innovating in IoT, crafting smart home solutions, or deploying complex industrial controls, the **IoTsmart ESP32-S3** connects seamlessly with **IoTextra** series expansion boards (mezzanines) via a 12-pin HOST connector. Essentially, this module upgrades standard mezzanines into intelligent, programmable smart modules.

The module offers two orientations for the **HOST** connector:

- **H-HOST** (horizontal connector), mounted on the **top-side** of the module, designed for **vertical** mezzanine installation
- **V-HOST** (vertical connector), mounted on the **bottom-side** of the module, designed for **horizontal** mezzanine installation

IoTsmart ESP32-S3 module (**top-side**)

IoTsmart ESP32-S3 module (**bottom-side**)

*Note: Some mezzanines, such as certain versions of the **IoTextra Relay2**, are too tall for the **V-HOST** connector.*

The module is equipped with two [Qwiic©](#) connectors for **I²C** communication and supports optional **EYESPI** (for connecting displays). **UART** connector is supported but must be installed by the user.

An onboard EEPROM (8 Kbit or 16 Kbit) stores configuration and user data and is accessible via **I²C** at addresses 0x54–0x57 by default.

The module incorporates **Waveshare ESP32-S3-Mini**, which provides:

- ESP32-S3FH4R2 chip with Xtensa 32-bit LX7 dual-core processor, up to 240MHz main frequency
- Built in 512KB of SRAM and 384KB ROM, onboard 2MB PSRAM and 4MB Flash memory
- Integrated 2.4GHz Wi-Fi and Bluetooth LE dual-mode wireless communication, with superior RF performance
- FPC 8-pin connector for USB
- USB Type-C interface via the **Tiny Adapter Board**

The module is powered by a +5VDC input.

The **IoTsmart ESP32-S3** module measures 47 x 30 mm.

Common applications:

Application	Target IoTextra Module	Software Stack
Industrial Hydro-monitoring	**IoTextra Analog** (0-10V и 4-20mA) / **EYESPI TFT Display**	LVGL (Light and Versatile Graphics Library / IoTflow / Node-RED
Building Automation	**IoTextra SSR Small / IoTextra Relay2**	Tasmota / MQTT
Edge Logic & Control	**IoTextra Octal2** (Isolated 4DI + 4DO)	IoTflow / Blynk
Smart Agriculture	**IoTextra Combo** (AI + Relay)	Blynk / IoTflow
Energy & Infrastructure	**IoTextra Input** (Isolated 8DI)	Tasmota / ESPHome

QUICK START

The **IoTsmart ESP32-S3** module is most frequently used with **IoTextra** series mezzanine. This combination allows the **IoTsmart ESP32-S3** module and a mezzanine to easily function as a standalone smart device.

The photos below illustrate the **IoTsmart ESP32-S3** module paired with various mezzanines and demonstrate both horizontal (**H-HOST**) and vertical (**V-HOST**) installation of the **HOST** connector:

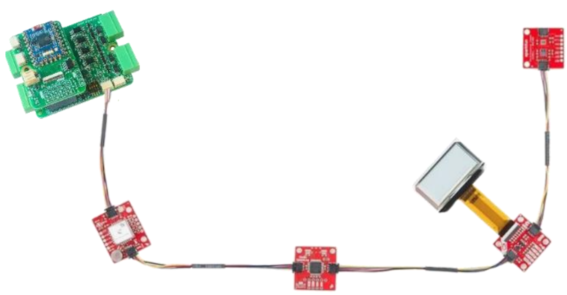
IoTsmart ESP32-S3 with **IoTextra Octal2**

IoTsmart ESP32-S3 with **IoTextra Input**

IoTsmart ESP32-S3 with **IoTextra SSR Small**

IoTsmart ESP32-S3 with **IoTextra Analog**

The **IoTsmart ESP32-S3** also readily connects to numerous [Qwiic©](#) compatible sensors, peripherals and modules via the **I²C** connector:



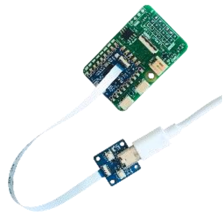
WAVESHARE ESP32-S3-MINI

The **IoTsmart ESP32-S3** module utilizes the **[Waveshare ESP32-S3-Tiny]**, which incorporates Xtensa 32-bit LX7 dual-core processor, up to 240MHz main frequency, 512KB of SRAM and 384KB ROM, onboard 2MB PSRAM and 4MB Flash memory.

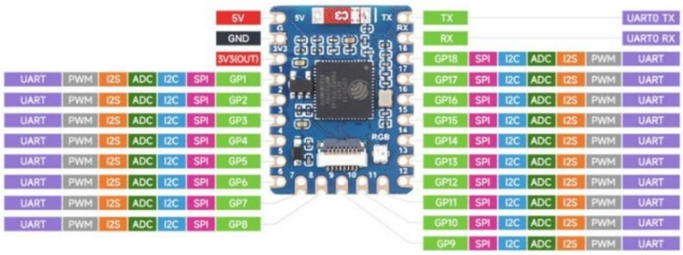
ESP32-S3 microcontroller integrated 2.4GHz Wi-Fi and Bluetooth© LE dual-mode wireless communication, with superior RF performance.

It also includes an on-board FPC 8-pin connector, which adapts the USB Type-C port via the **Tiny Adaptor Board**.

The following photo displays the **Tiny Adaptor Board** with a cable; however, these are not included with the **IoTsmart ESP32-S3** and must be purchased separately:



For your information, here is the pinout of the **Waveshare ESP32-S3-Tiny**:

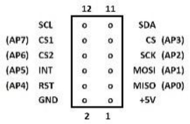


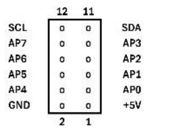
CONNECTORS

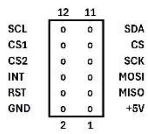
The module is equipped with the following connectors:

- **H-HOST** - horizontal (right-angle) connector installed on the **top-side**, designed for **vertical** mezzanine installation, **V-HOST** - vertical connector installed on the **bottom-side**, designed for **horizontal** mezzanine installation. Information regarding the type of the installed **HOST** connector is provided on the **bottom-side** of the module
- **EYESPI** connector is intended for communication with displays (indicators) using an 18-pin FFC cable.
- Two **Qwiic©** connectors for connecting external sensors and devices via the **I²C** bus.
- An optional **UART** connector (not pre-installed)
- A **PWR** connector (+5VDC input)

HOST connector. Pinout of the **HOST** connectors:



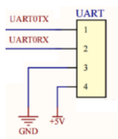




HOST on the IoTsmart module	HOST-P12 on the mezzanine	HOST-S on the mezzanine

The **HOST** connector of the **IoTsmart ESP32-S3** module is inserted into the **HOST-12** or **HOST-S** connector on the mezzanine. Therefore, the pinout for the **HOST-P12** and **HOST-S** connectors on the mezzanines is also shown in the figure to compare the connector signals.

UART connector. The contacts of this connector are shown in the figure:



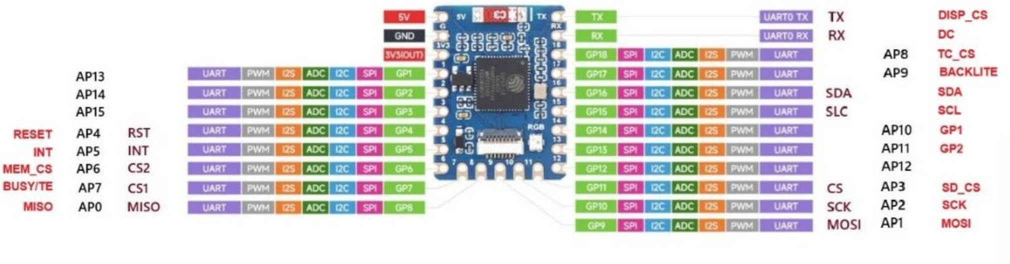
EYESPI Connector. The module's top side has an **EYESPI** connector. It is intended for communication with displays (indicators) using an 18-pin FFC cable. Communication with indicators is achieved due to compatibility with the 18-pin **EYESPI Cable** used by **Adafruit**. The standardization of this cable allows the following displays to be connected:

- [2.0" 320x240 Color IPS TFT Display with microSD Card Breakout]
- [2.2" 18-bit color TFT LCD display with microSD card breakout]
- [1.8" Color TFT LCD display with MicroSD Card Breakout]
- [OLED Breakout Board - 16-bit Color 1.5" w/microSD holder]
- [OLED Breakout Board - 16-bit Color 1.27" w/microSD holder]
- [Adafruit 2.7" Tri-Color eInk]
- [Adafruit 1.54" Monochrome eInk]

[Waveshare 2" Capacitive Touch Display Module] can also be connected using an 18-pin FFC cable.
Here is the correspondence between the module's signals and the signals in the **EYESPI** connector:

EYESPI		Signals on the IoTsmart ESP32-S3 module schematic
pin #	**NAME**	
1	VCC	+3V3
2	BACKLITE	AP9
3	GND	GND
4	SCK	AP2
5	MOSI	AP1
6	MISO	AP0
7	DC	RX
8	RESET	AP4
9	DISP_CS	TX
10	CD_CS	AP3
11	MEM_CS	AP6
12	TC_CS	AP8
13	SCL	SCL
14	SDA	SDA
15	INT	AP5
16	GP1	AP10
17	GP2	AP11
18	BUSY/TE	AP7

The **HOST** and **EYESPI** **connector** signals correspond to the **Waveshare ESP32-S3-Tiny** signals as follows:



EEPROM

To store configuration and other user information, the **IoTsmart ESP32-S3** module includes an onboard **EEPROM** (8 Kbit or 16 Kbit). This **EEPROM** is accessible via the I^2C bus and is visible at addresses 0x54-0x57 by default.

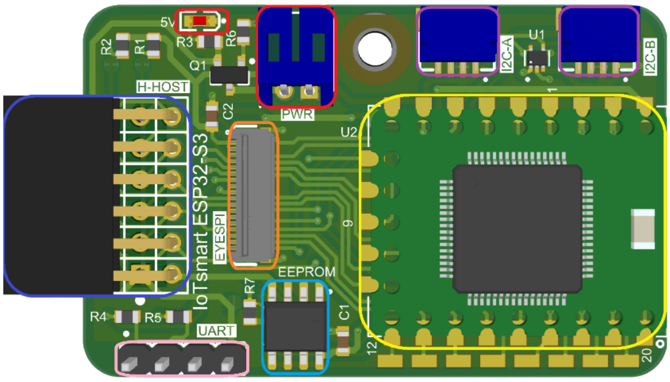
POWER SUPPLY

The module operates from a 5VDC power input supply. The power supply can be connected via a 2-pin **PWR** connector (JST S2B-PH-K-S, 2.00 mm pitch) or through the **Tiny Adapter Board**.

The typical power consumption of the module with the **Waveshare ESP32-S3-Tiny** is approximately 25 mA (measured without additional connected peripherals or expansion boards). The maximum current consumption, including power provided to a mezzanine, should not exceed 1000 mA.

LAYOUT

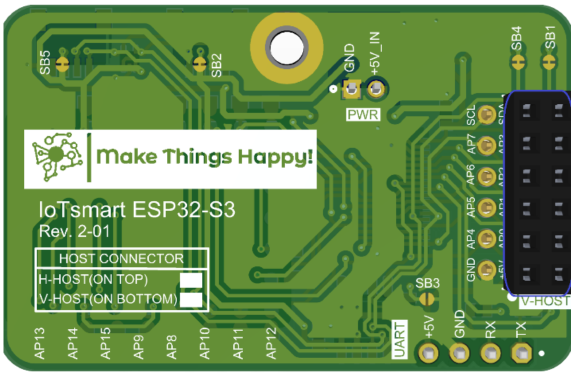
Below is the layout of the elements on the **top-side** of the **IoTsmart ESP32-S3** module when using the **HOST-H** connector:



In this picture:

- Power-related elements (external power connector, named **PWR**, and **LED**) are highlighted in **red**
- The soldered **Waveshare ESP32-S3-Tiny** is highlighted in **yellow**
- The **HOST-H** connector is highlighted in **blue**
- The **EYESPI** connector is highlighted in **orange**
- **UART** connector not installed during production are highlighted in pink
- The **Qwiic©** connectors for connecting peripherals via the **I²C** bus is highlighted in **purple**
- EEPROM is highlighted in light blue

Below is the layout of the **IoTsmart ESP32-S3** module **bottom-side** when using the **HOST-V** connector (highlighted in **blue**):



JUMPERS

Jumpers are located on the underside of the module:

- SB1, SB4: Connect pull-up resistors to **I²C** SCL and SDA (enabled by default)
- SB2, SB5: Disables +3.3V power to **Qwiic©** devices when open (default: closed)
- SB3: Sets EEPROM **I²C** address

b7	b6	b5	b4	b3	b2	b1	b0
1	0	1	0	1	x	x	R/W

CONFIGURATION TABLES

The **bottom-side** of the module provides information about the type of **HOST** connector installed:

- **H-HOST** - horizontal (right-angle) connector installed on the **top-side**, designed for **vertical** mezzanine installation.
- **V-HOST** - vertical connector installed on the **bottom-side**, designed for **horizontal** mezzanine installation.

COMPATIBILITY WITH MEZZANINES

The **IoTsmart ESP32-S3** is compatible with all **IoTextra** mezzanines, including

Digital IoTextra	Analog IoTextra	Combo IoTextra
• IoTextra Input • IoTextra Relay2 • IoTextra SSR Smal • IoTextra MOSFET2 • IoTextra Octal • IoTextra Octal2	• IoTextra Analog • IoTextra Analog2 • IoTextra Analog3	• IoTextra Combo • IoTextra Combo2

ACCESSORIES

Recommended accessories:

- Two-pin cable with a JST connector (for Power Supply)
- **Tiny Adapter Board** with USB Type-C
- **Qwiic© I²C** cable (connector on both ends)
- 18-pin FPC standard **EYESPI** cable for connecting a display (indicator) or inter-module connection, 100mm or 200mm long